



TECHNICAL AND SCIENTIFIC WRITING

YEAR- II (SEMESTER- III)

GROUP ASSIGNMENT

**CHANGING DEMOGRAPHIC TRENDS IN INDIA AND ITS
IMPLICATIONS**

SUBMITTED TO-

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INTRODUCTION

India is at a point where it is undergoing major demographic shifts causing a certain speculation for economic and social implications for the nation. India's demographic issues are nothing short of complex. A fall in fertility with a simultaneous rise in life expectancy are setting us on the course of becoming an ageing population. There is an imbalance among states, with northern states' maintaining an ever-high fertility as compared to their southern counterparts, "replacement migration" is on a rise to fill the labour gap in such states. Despite of these challenges, India is seeing a surge in its working-age population, a trend which is to continue for a while in the near-future. This indicates larger labour supplies and gains in per capita income, positively contributing to the country's demographic dividend. The shifts are not limited to these macro scenarios alone, for the effects of this shift is pertinent and loud in family structures, urbanisation patterns, healthcare, monetary-welfare policies for target age-groups, migration etc. This report aims to understand these complex, inter-vined issues and understand the nuances of by thematically segregating and subsequently, critically analysing them.

METHODOLOGY

At the primary-level, all the group members went through different research papers to educate ourselves with the bases of the chosen topic. A total of 10 research papers were studied for this report. After conducting this literature review, major recurring themes were identified, chosen, and assigned in the format- "one theme for each member". A unique colour was assigned for each theme:

1. Changing Demographic Pattern – **Blue**
2. Demographic Dividend and Economic Consequences – **Purple**
3. Trends in Fertility– **Orange**

This approach was aimed at streamlining the annotating process for all the group members.

THEMATIC SYNTHESIS

Theme I: Changing Demographic Pattern

After reading the articles and highlighting the theme changing demographic trends, one can see that India is experiencing a significant demographic change with several level to it. The most highlighted fact in the readings is that India is seeing a rapid change in its age structure, fertility is decreasing, life expectancy is increasing and this trend has been altering the whole population structure. It has been noted in the papers that the proportion of the working age population has been on the rise in the past few decades to cause what is popularly known as the demographic dividend. A number of studies assume that this age structural change has already had beneficiallyeffected the economy of the country, but that the degree of benefit is also dependent on policy support and labour market conditions.

The other theme that is brought out is that the demographic change is not evenly spread throughout the nation. A few states notably those in the south have seen rapid decline in the fertility rate and have already gone into places of population ageing. On the other handthe states in the north and central states also maintain younger populations and will continue to supply a significant portion of the future labour force of India. Due to this imbalance, the demographic transition is posing varied opportunities and challenges in various regions. Few articles showed that in the absence of proper jobs, skills and participation of gender in the labour market, mere fact that there is a young population will not necessarily translate to economic growth.

One of the trends that can be seen throughout the literature is the gradual increase in the number of the elderly. Some articles indicate that older adults will greatly increase in the number of districts within India by 2030 and 2040 and will strain the social security, health services and care giving structures. The readings also demonstrated that the rate of ageing in urban and highly developed areas is accelerating and the rural areas are trailing. This causes a demographic disparity both on a state level and even a district level.

Another topic covered in the papers is the fact that the demographic change is linked to the process of urbanisation and family changes. The fertility and life expectancy in urban areas are reduced and this increases the speed of the demographic transition. Meanwhile in urban areas families are reducing in size, the household is getting smaller and the traditional support

system is evolving gradually. The trends have a direct influence on the nature of the demographic patterns particularly the dependency ratios, care giving roles and the patterns of marriages.

Altogether, all the literature demonstrates that Indian demographic transition is a multifaceted process on the one hand, it opens opportunities due to the large numbers of workingage population and on the other hand, it shows certain challenges that are associated with the ageing of the population, regional differences, labour market mismatch and family transformations. The trends emphasized throughout the papers imply that India needs to be ready to live in the future when its population will not be evenly youthful and the demographic variations will play an ever growing role in the social, economic and policy decisionmaking.

Theme II: Demographic Dividend and Economic Consequences

To any economy, its population is its backbone, supporting and enabling its survival, and a demographic shift of any kind will impact the economic well-being of the society. This aftermath has been studied in accomplished academic papers through conceptualizing and quantitative measures viewing events from the lenses of urbanisation, work-age population, geographical comparison of states, employment, statistical indicators etc. The demographic shift in acting like a major economic force reshaping labour supply, dependency ratios etc. The research consistently points out a growing working-age population finding that India's per capita income has lied between 0.3-1 percentage point annually subject to the time and the state but. However, a severe north-south divide has been identified in terms of replacement level fertility which directly impacts the future-availability of working-population in southern states. It is argued that this number is less in especially western and southern states while the northern states continue to thrive. Such circumstances are instigating instances of replacement migration to fulfil that gap. Though, another flip-side to this is infrastructural, governance and educational differences across these states. The southern and western states derive high value from a positive demographic dividend, however, the northern states find it hard to keep up with the accumulating labour supply for they are not able to generate enough economic activities to absorb this supply. These shifts are also felt at the micro-level as demographic ageing brings about excessive economic burden due to more old-age dependent people, increasing healthcare costs and weakened intergenerational support from the family due to today's life choices and smaller family size.

There exist both, challenges and opportunities as a result of this demographic shift, what is required of us is to maintain a balance these two and ensure reaping the boon of a larger working age population, while promoting the maintenance and balance of replacing rate fertility to tackle the challenges of an increased dependent population. In more general terms, the policy responsiveness, institutional strength, and labour-market adaptability are more important to India's economic trajectory right now than population dynamics alone.

Theme III: Trends in Fertility

Fertility is highlighted as a key theme in the context of India's demographic transition. If we look through the information from the literature, it is seen that while India has achieved near replacement-level fertility, which by definition is the average number of children per woman required to replace one generation to the next, the transition has not been uniform. The factors that have contributed to it are the disparities across regions, policy-driven interventions, and cultural measures or behaviours that have deeply shaped it. At the same time, phenomena such as son preference, skewed sex ratios, and constrained female autonomy reveal that gender is not a secondary but a central force shaping demographic behaviour.

The literature collectively characterises India's decline in fertility as fast-paced, with levels falling from over five children per woman in the 1970s to 2.2 presently, and it is further projected to fall below the replacement level of 2.1 within the next decade. Moreover, fertility decline is heterogeneous, showing profound spatial disparities. Urban TFR (1.9) is significantly lower than rural TFR (2.7) in 2011. A stark difference is seen between northern and southern states as well. Southern states such as Kerala, Tamil Nadu, and Andhra Pradesh have long maintained below-replacement fertility, while large northern states such as Uttar Pradesh, Bihar, Rajasthan, and Madhya Pradesh exhibit persistently higher fertility. The disparity in rural and urban areas, as well as northern states, reflects how deeply entrenched cultural norms, limited autonomy of women and unequal development also affect the fertility rate. For instance, fertility decline has intensified son preference rather than reduce it. In a situation where families desire one or two offspring, the pressure for at least one male offspring increases. As a consequence, among poorer households, female child mortality occurs due to partial care and in urban households, sex-selective abortions occur.

Throughout the literature, urbanisation is increasingly associated with a decline in the fertility rate. Urbanisation, improved access to education and healthcare correlate with better access to sexual and reproductive health services that allow for more effective regulation of fertility.

In urban areas, women generally want smaller families, marry late, and use contraceptives more frequently and possibly more efficiently.

Changes in social indicators such as education and an increase in women's employment have also affected fertility levels. For example, in Kerala, the high decline in fertility levels was associated with high rates of female literacy.

India does not follow the conventional pattern of demographic transition; policy decisions have played a pivotal role in the decline of fertility levels in India. Concentrated government efforts to promote family planning have led to a substantial decline in fertility. This was achieved primarily through the use of contraceptives, almost solely using female sterilisation. However, this has a drawback. Sterilisation is happening mostly at young ages, and this irreversible procedure may prove detrimental to women's health as a whole. Promoting reversible methods of contraception is key to ensuring healthy birth spacing, healthier mothers and offspring, as well as a sustainable pattern for fertility decline.

Looking ahead, this prolonged decline in fertility will lead to rapid population ageing, with the elderly population expected to triple in the coming decades. The ultimate outcome of whether this decline in fertility will benefit India depends heavily on its ability to leverage its demographic dividend.

CONCLUSION

The thematic synthesis further strengthens the position of structural-complexity faced by the Indian society as a result of the recent, but major demographic shifts. Though, the increase in "working-age population" can prove to be a significant asset for the socio-economic rise and stability of the nation, this phenomenon will not last forever. The simultaneous decrease in fertility rates forces us to consider a future where India's current position of "the youngest population" may be reversed. This would mean a rapid rise in the share of elderly (dependant) population, which directly points to a fall in the labour force. If such a situation is to arise, it will be detrimental for people across age brackets, for a lack of labour force directly translates to lesser qualified professionals in all industries. Conclusively, we can say that India needs to understand the prime work-force it entails right now and maximise the return on that, but it is to be remembered that policy nudges are required to keep a viable

fertility rate across the country to maintain a positive demographic dividend, as well as ensuring sound social security for the population.

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CONTRIBUTION SECTION

- **All Members** – Reading; Annotating; Writing; Methodology
- **Ravi Pratap Singh**– Analysing the changes in the population structure over time and subsequent social consequences under the Theme – “Changing Demographic Pattern”
- **Satvik Bhardwaj** – Introduction; Thematic analysis under the Theme –“Demographic Dividend and Economic Consequences”; Conclusion; Formatting of the Report; Compilation
- **Vidita Ganesh** – Analysing Fertility and related trends and thematic analysis of the same across articles under the Theme- “Trends in Fertility”